MARS-O-RAN: Mission-Aware Risk-Constrained Elastic Network Slicing for Emergency Robotics, with Statistical and Edge-Case Evaluation

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***Abstract*—When robots respond to an industrial accident, the radio link that matters most shifts from one minute to the next. In our gas-leak scenario, five dependent steps (detecting the gas, localizing its source, driving a ground robot to the valve, inspecting the valve, and closing it) each impose their own deadline, data-freshness, and reliability target on a shared cell. We introduce MARS-O-RAN, a digital twin in the style of an Open Radio Access Network (O-RAN) that splits physical resource blocks (PRBs) among four slices according to mission progress and risk. Four policies are compared: STATIC, PROPORTIONAL, MISSION-AWARE, and the proposed MARS controller. Over 30 paired random-seed runs per policy, both mission-oriented policies raise the task-level mission-success score from 0.187 (STATIC) to above 0.70 and cut mean mission loss by about 26 units in every paired seed (Wilcoxon $p < 10^{-5}$). Relative to MISSION-AWARE, MARS reduces reconfiguration cost by 43.7% in all 30 seeds, while its success score (0.707 vs. 0.727) and $\mathrm{CVaR}_{0.90}$ loss (23.3 vs. 24.5) are statistically indistinguishable. We further stress the controllers with 11 edge-case studies covering PRB budgets from the feasibility limit to twice nominal, incident severity, network-shock rate, deadline tightness, workload scaling, controller parameters, and an extended 30-seed ablation. The edge cases reveal that MARS gains up to 8.7 percentage points under mild incidents but loses up to 16 points when slack is tight, that the final valve-manipulation task is infeasible under the nominal reliability model for every policy, that additional PRBs can reduce success through radio–edge coupling, and that the criticality weights are inert because the allocation saturates at slice bounds. The study is a reproducible software simulation, not a field validation of 5G/6G or O-RAN equipment.**

***Index Terms*—O-RAN, network slicing, emergency robotics, mission-aware control, edge computing, Age of Information, CVaR, edge-case evaluation.**

I. Introduction

A ground robot closing a leaking valve, a drone streaming camera frames to an edge server, responders viewing hazard overlays, and a field of gas sensors can all share one private cell during an industrial emergency. Their demands conflict, and which one deserves the spectrum depends on where the mission stands. Camera uplink dominates while the leak is being located; a few kilobits of control traffic, delivered quickly and reliably, matter more once the robot grips the valve. A slicing policy that looks only at instantaneous traffic volume cannot see this shift.

Open RAN makes this kind of application-driven control architecturally plausible. Its RAN Intelligent Controllers run optimization loops at two speeds: a near-real-time controller that acts on the base station over E2, and a slower controller that issues policies over A1 [1]–[3]. A mission-aware slicing rule would naturally sit in the faster loop. We study the logic of such a rule inside a Python simulation; no E2 or A1 messages are exchanged and no RIC software is deployed.

We ask two questions. First, *can tying PRB shares to the currently executable mission step improve task completion over fixed or demand-driven slicing?* Second, *under which budgets, incident severities, deadlines, and workloads does that advantage disappear or reverse?* The second question receives most of our attention, because a controller tuned at one operating point says little about its behavior elsewhere.

Contributions. (1) A mission-task graph that links slice demand to task dependencies; (2) a heuristic allocation rule driven by deadline urgency, hazard, critical-path status, freshness, and risk, plus a risk reserve and stability constraints; (3) a joint evaluation model for radio service, edge computation, deadline, freshness, reliability, reconfiguration, and tail loss; (4) a statistically grounded comparison of four policies using paired seeds, bootstrap confidence intervals, and signed-rank tests; and (5) a systematic edge-case study that maps the operating envelope of each policy and identifies the model mechanisms responsible for failures. Contribution (5) is new in this version and changes several conclusions of the earlier draft, which we report explicitly.

II. Related Work and Background

Open RAN control. The O-RAN specifications separate a near-real-time controller from a non-real-time one [1], [2], and Polese *et al.* review the interfaces and the learning-based applications proposed for both [3]. Our simulator mirrors this split only loosely: scenario settings are fixed before a run, playing the role of a slow policy, and a per-second heuristic plays the role of the fast loop.

Slicing with guarantees. Much recent O-RAN slicing work optimizes service-level agreements with reinforcement learning. SafeSlice, for instance, adds a safety layer so that a deep RL agent respects both cumulative and per-step latency limits [9]. Baguer *et al.* show that slicing over Open RAN can sustain beyond-visual-line-of-sight drone operation [8]. Both target per-slice performance targets; neither models a sequence of dependent tasks whose value changes as the mission advances, which is the gap MARS-O-RAN explores.

TABLE I
MISSION TASKS (SYNTHETIC CONFIGURATION VALUES).

Task	Slice	D_k (s)	L_k (Mb)	C_k	w_k	h_k	$\epsilon_k^{\max}$
K_1 detection	SV	36	0.2	20	5	0.70	0.08
K_2 localization	SM	86	12.0	120	8	0.80	0.10
K_3 navigation	SC	125	1.2	55	10	0.95	0.06
K_4 inspection	SM	168	7.0	100	9	0.90	0.08
K_5 manipulation	SC	205	1.0	50	12	1.00	0.04

Freshness, short packets, and tail risk. We borrow three tools from adjacent fields. Age of Information quantifies how stale the newest delivered update is [4], [5]; a robot steering toward a valve is better served by a recent position than by a large but outdated batch. Finite-blocklength theory shows that short control packets cannot reach Shannon capacity at low error rates [6], which motivates the rate penalty in our radio model. Conditional value at risk summarizes the worst part of a loss distribution and can be optimized through a convex reformulation [7]; we use it only to score simulated outcomes. Validating our abstractions with a packet-level simulator such as 5G-LENA [10] remains future work.

III. SCENARIO AND SYSTEM MODEL

A. Network and Slices

The digital twin models one cell and one edge server. The radio budget is $B = 100$ PRBs in the nominal scenario. Let $b_s(t)$ be the PRBs given to slice $s \in \mathcal{S} = \{\mathrm{SC}, \mathrm{SV}, \mathrm{SX}, \mathrm{SM}\}$ (robot control, perception video, responder XR, safety sensing) at step t (1 s). The allocation must satisfy

$$\sum_{s\in\mathcal{S}} b_s(t) \le B, \qquad B_s^{\min} \le b_s(t) \le B_s^{\max}, \tag{1}$$

with $B^{\min} = (15, 15, 10, 10)$ and $B^{\max} = (60, 60, 50, 40)$. Hence $\sum_s B_s^{\min} = 50$ is the smallest feasible budget and $\sum_s B_s^{\max} = 210$ the largest usable one; both limits are probed in Section VII.

B. Mission Task Graph

The mission is a chain-structured DAG $\mathcal{G} = (\mathcal{K}, \mathcal{E})$ with tasks K_1 gas detection (SV), K_2 source localization (SM), K_3 UGV navigation (SC), K_4 valve inspection (SM), and K_5 valve manipulation (SC). A task may start only after its predecessor finishes, and only one task executes at a time. Each task k has deadline D_k, bit workload L_k, compute workload C_k, value w_k, hazard weight h_k, freshness limit $\Delta_k^{\max}$, and failure budget $\epsilon_k^{\max}$ (Table I).

C. Incident, Radio, and Edge Model

The incident starts at $t_0 = 20$ s. Severity is $\sigma(t) = 0.05$ before t_0 and $\sigma(t) = \mathrm{clip}(\sigma_0 + 0.0015(t - t_0) + n_t, 0, 1)$ afterwards, with $\sigma_0 = 0.6$ and $n_t \sim \mathcal{N}(0, 0.025^2)$. A network shock of 5 s starts with probability $p_{\mathrm{sh}}(1+\sigma)$ per step ($p_{\mathrm{sh}} = 0.018$) and degrades channel, edge, and backhaul factors. The service rate of the active task's slice is

$$R_s(t) = b_s(t)\, W_{\mathrm{PRB}} \log_2(1 + \gamma_s(t))\, \eta\, h_s(t)\, F_s^{\mathrm{FBL}}, \tag{2}$$

with $W_{\mathrm{PRB}} = 180$ kHz, $\eta = 0.62$, channel factor h_s, and a finite-blocklength correction $F_s^{\mathrm{FBL}} \in [0.55, 1]$. Edge capacity $C_{\mathrm{edge}} = 100$ units is shared in proportion to radio share,

$$f_s(t) = C_{\mathrm{edge}} \frac{b_s(t)}{B} \left(1 + 0.15\, \mathbf{1}[s = s_{\mathrm{act}}]\right), \tag{3}$$

renormalized to C_{edge}; the 15% bonus applies only to the mission-oriented policies. Equation (3) couples radio and edge decisions, which matters for the budget edge case in Section VII. A task finishes when both its remaining bits and its remaining compute reach zero.

D. Freshness, Reliability, and Task Success

AoI resets to 1 s when a slice receives at least its minimum (plus 5 PRBs for SC/SV) and otherwise increases by 1 s. End-to-end reliability is the product of radio, edge, and backhaul terms,

$$\begin{aligned} P^{\mathrm{succ}} &= P^{\mathrm{radio}}\, P^{\mathrm{edge}}\, P^{\mathrm{bh}}, \\ P^{\mathrm{radio}} &= 0.999 - 0.030\sigma - \ldots, \\ P^{\mathrm{edge}} &= 0.997 - 0.025\sigma - \ldots, \\ P^{\mathrm{bh}} &= 0.998 - 0.020\sigma - \ldots, \end{aligned} \tag{4}$$

where the omitted terms are load- and shock-dependent penalties. Task k succeeds ($q_k = 1$) only if it meets its deadline, its slice AoI is within $\Delta_k^{\max}$, and $P^{\mathrm{succ}} \ge 1 - \epsilon_k^{\max}$.

E. Metrics

The *mission-success score* is the number of successful tasks divided by five (unobserved tasks count as failures). The achieved value of a task is w_k on success, $0.45w_k$ if only the deadline was met, and $0.10w_k$ otherwise; the *mission loss* is

$$L = \sum_k w_k(1 + 0.25h_k) - \sum_k (\text{achieved value})_k, \tag{5}$$

which ranges from 9.875 (all tasks succeed) to 53.875 (no task observed). The reconfiguration cost is $\sum_t \sum_s |b_s(t) - b_s(t-1)|$. Tail risk is

$$\mathrm{CVaR}_\beta(L) = \min_\eta \left[\eta + \tfrac{1}{1-\beta}\mathbb{E}[(L-\eta)^+]\right], \tag{6}$$

estimated empirically as the mean of losses at or above the β-quantile across seeds [7].

IV. POLICIES AND THE MARS CONTROLLER

A. Dynamic Mission Criticality

For the executable task k,

$$\chi_k(t) = w_k\left[1 + \alpha_u U_k + \alpha_h H_k + \alpha_{cp} I_{cp,k} + \alpha_a A_k + \alpha_r R_k\right], \tag{7}$$

where $U_k = \min\{1, 40/(D_k - t + 2)\}$ is normalized urgency, A_k is normalized AoI pressure, $I_{cp,k} \in \{0.35, 0.6, 1\}$ marks critical-path position, and $H_k = R_k = \sigma h_k$. Note that in the current implementation the hazard and risk terms are the same quantity, so α_h and α_r act as a single coefficient.

Algorithm 1 One MARS control step

observe $\sigma(t)$, shock flag, demand d_s, previous $b_s(t-1)$
$k \leftarrow$ executable task in $\mathcal{G}$; $u_s \leftarrow 0.35\, d_s / \sum d$; $u_{s(k)} \mathrel{+}= \chi_k(t)$
$r \leftarrow$ risk reserve$(\sigma, \text{shock}, \rho)$; $u_{\text{SC}} \mathrel{+}= 0.45r$; $u_{\text{SV}} \mathrel{+}= 0.35r$; $u_{s(k)} \mathrel{+}= 0.30r$
$u_s \mathrel{-}= \lambda\,|b_s(t-1) - 25|$; $\omega \leftarrow \text{softmax}(u)$
$b_s \leftarrow \text{clip}(\text{round}(B\omega_s), B_s^{\min}, B_s^{\max})$
if $t - t_{\text{last}} < T_h$ **then** $b \leftarrow b(t-1)$
else $b_s \leftarrow \text{clip}(b_s,\, b_s(t-1) \pm \Delta_{\max})$
end if
repair $\sum_s b_s = B$ (remove from largest surplus, add by unmet demand)
serve radio and edge workloads; update AoI, reliability, task state

TABLE II
EDGE-CASE STUDIES (30 SEEDS PER POINT).

Study	Values	Edge case probed
PRB budget B	50–200	feasibility floor; surplus
Initial severity σ_0	0–1.0	benign vs. extreme incident
Shock prob. p_{sh}	0–0.30	no shocks vs. shock storm
Deadline scale	0.5–1.5	infeasible vs. relaxed deadlines
Workload scale	0.25–3.0	light load vs. horizon overrun
Severity $\times$ B grid	5×4	joint stress
Risk aversion ρ	0–1	MARS reserve strength
Hold time T_h	0–20 s	hysteresis off vs. sluggish
Change limit $\Delta_{\max}$	1–64	frozen vs. unconstrained
Ablation (8 variants)	—	component contribution
Weight logging	—	softmax saturation

B. Allocation Policies

STATIC uses fixed shares $(0.29, 0.26, 0.18, 0.27)$. PROPORTIONAL uses demand shares. MISSION-AWARE scores each slice by χ_k of its executable task plus 0.35 times its demand share, adds an 18% bonus to the best slice, and maps scores to weights with a softmax. MARS replaces the bonus with (i) a severity-dependent risk reserve added to SC (45%), SV (35%), and the target slice (30%), scaled by risk aversion ρ; (ii) a penalty proportional to the deviation of the previous allocation from the equal share of 25 PRBs; and (iii) a hold time $T_h = 4\,\text{s}$ during which the allocation is frozen. All policies are limited to $\Delta_{\max} = 16$ PRBs of change per slice per step. Algorithm 1 summarizes one MARS step.

The controller is a deterministic heuristic; it does not solve the underlying mixed-integer stochastic program, and CVaR is evaluated after the fact rather than enforced online.

V. Experimental Method

Main experiment. Each policy is run for 300 s on seeds 20–49 (120 runs). Because the stochastic environment draws the same random numbers regardless of the policy, a given seed exposes all four policies to an identical severity, shock, channel, and demand trajectory. We verified this property programmatically and exploit it with paired tests (common random numbers). Re-running the experiment reproduces the archived results exactly.

Statistics. We report 95% percentile bootstrap confidence intervals (4000 resamples) for means and for $\text{CVaR}_{0.90}$ [11], and paired Wilcoxon signed-rank tests on per-seed differences [12]. With 30 seeds, the $\text{CVaR}_{0.90}$ estimate averages the worst three runs, so its interval is wide.

Edge-case studies. Table II lists the sweeps. Each point uses all four policies (or MARS alone for controller parameters) and the same 30 seeds, about 8,000 runs in total. Deadline scaling stretches each deadline relative to the incident start, $D'_k = t_0 + s(D_k - t_0)$; workload scaling multiplies L_k and C_k.

VI. Results at the Nominal Operating Point

A. Aggregate Performance with Uncertainty

Table III and Fig. 1 report the main results with 95% confidence intervals. Fig. 2 shows the four success components. Both mission-oriented policies achieve every deadline and quadruple the success score of STATIC and PROPORTIONAL. MARS has the lowest point estimate of $\text{CVaR}_{0.90}$, but its interval [22.0, 26.9] overlaps MISSION-AWARE's [20.9, 29.7].

B. Paired Comparisons

Table IV tests MARS against each baseline seed by seed. Against STATIC and PROPORTIONAL, MARS lowers mission loss in all 30 seeds. Against MISSION-AWARE, MARS lowers reconfiguration cost in all 30 seeds (mean -799 PRB-steps). Mission loss and success are tied in 17 seeds, better in 5, and worse in 8, and neither difference is significant. *The defensible claim is therefore equivalent mission outcome at 44% lower reconfiguration cost, not superior tail risk.*

C. Loss Distribution and Choice of β

Fig. 4(a) shows the per-run loss distribution. The mission-oriented policies concentrate near 16.5 (four of five tasks succeed), and the non-adaptive policies concentrate at 45.0 (only K_1 succeeds). Fig. 4(b) plots CVaR_β for $\beta \in [0, 0.95]$. MARS has a higher mean ($\beta = 0$) than MISSION-AWARE and a higher CVaR for $0.55 \le \beta \le 0.85$ and $\beta = 0.95$; it is lower only near $\beta = 0.90$. The earlier draft's $\text{CVaR}_{0.90}$ advantage is therefore sensitive to the chosen confidence level, and we no longer present it as a finding.

D. Per-Task Anatomy and an Infeasible Task

Fig. 5(a) breaks success down by task. STATIC and PROPORTIONAL complete K_1 and then miss the deadline of K_2. Each later task inherits the delay, so 80% of their task instances fail on deadlines (Fig. 5(b)). The mission-oriented policies meet every deadline, and their failures are reliability failures only.

The most important edge case is K_5: *no policy completes valve manipulation successfully in any of the 120 runs.* K_5 requires $P^{\text{succ}} \ge 0.96$. Ignoring load and shock penalties, (4) gives $P^{\text{succ}} \approx 0.994(1 - 0.075\sigma)$, which falls below 0.96 once $\sigma \gtrsim 0.46$. With $\sigma_0 = 0.6$ and positive growth, severity during K_5 is about 0.8, so K_5 is infeasible by construction for every policy. The full-mission success probability is consequently zero, and the success score cannot exceed 0.8 at the nominal point. This is a property of the scenario parameters, not of the controllers. It is confirmed by the severity sweep, where success reaches 0.96 only when σ_0 is near zero (Fig. 10).

TABLE III

MAIN RESULT, 30 SEEDS PER POLICY, MEAN [95% BOOTSTRAP CI]. FULL-MISSION PROBABILITY IS THE FRACTION OF RUNS IN WHICH ALL FIVE TASKS SUCCEED.

Policy	Mission-success score	Deadline success	Reconfig. cost	Mean loss	$\text{CVaR}_{0.90}$ loss
STATIC	0.187 [0.167, 0.200]	0.200	14	45.16 [44.98, 45.43]	45.16 [44.98, 47.73]
PROPORTIONAL	0.193 [0.180, 0.200]	0.193	3414 [3389, 3438]	45.13 [44.98, 45.43]	45.13 [44.98, 49.48]
MISSION-AWARE	0.727 [0.680, 0.767]	1.000	1829 [1805, 1851]	18.07 [17.14, 19.19]	24.54 [20.88, 29.68]
MARS	0.707 [0.660, 0.747]	1.000	1030 [986, 1075]	18.64 [17.63, 19.78]	23.26 [21.98, 26.93]

Full-mission probability is 0.00 for every policy (see Section VI-D).

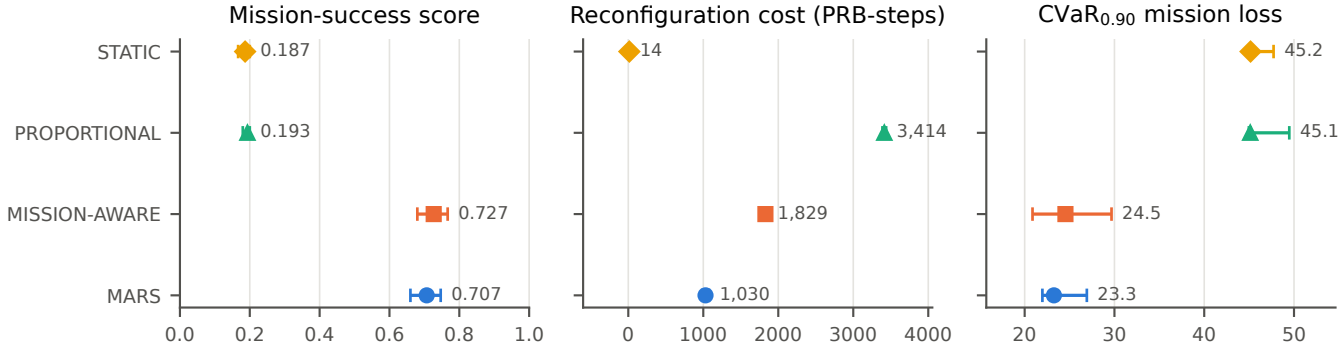


Fig. 1. Main metrics with 95% bootstrap confidence intervals (30 seeds). STATIC's reconfiguration cost of 14 is the one-time move from the equal-share initial allocation.

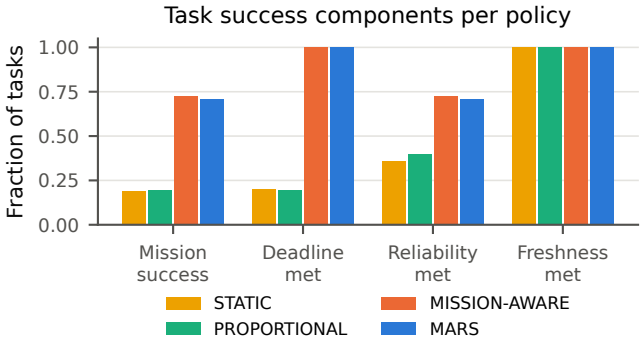


Fig. 2. Mean success components per policy. Freshness is satisfied in every run, so failures come from deadline or reliability checks.

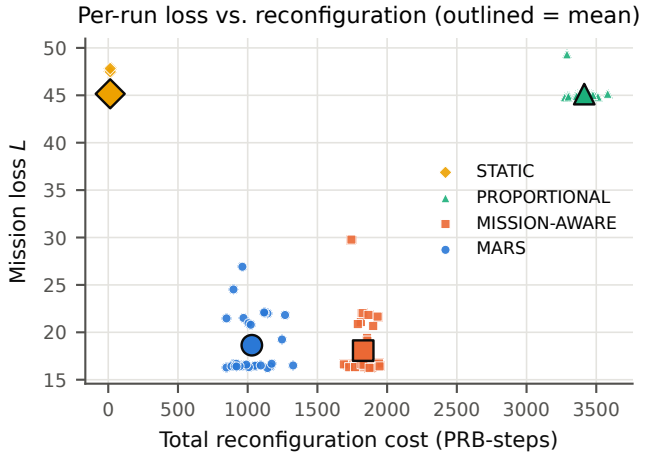


Fig. 3. Per-run mission loss versus reconfiguration cost (jitter ± 0.25 on loss). MARS and MISSION-AWARE overlap in loss but separate cleanly in cost; STATIC and PROPORTIONAL sit on the high-loss plateau.

TABLE IV

PAIRED DIFFERENCES MARS − BASELINE OVER 30 SEEDS: MEAN [95% CI], WIN/TIE/LOSS COUNT FOR MARS, WILCOXON SIGNED-RANK p.

Baseline	Metric	Δ mean [CI]	W/T/L	p
STATIC	loss	-26.5 $[-27.5, -25.4]$	30/0/0	$< 10^{-5}$
PROPORT.	loss	-26.5 $[-27.5, -25.4]$	30/0/0	$< 10^{-5}$
PROPORT.	reconfig.	-2384 $[-2432, -2334]$	30/0/0	$< 10^{-5}$
MISSION-AW.	loss	$+0.57$ $[-0.55, 1.65]$	5/17/8	0.23
MISSION-AW.	success	-0.020 $[-0.067, 0.027]$	5/17/8	0.47
MISSION-AW.	reconfig.	-799 $[-835, -762]$	30/0/0	$< 10^{-5}$

E. Allocation Dynamics

Fig. 7 contrasts MISSION-AWARE and MARS on seed 20. Both move the active slice to its upper bound (60 PRBs for SC/SV, 40 for SM). MISSION-AWARE jitters the non-active slices every step, whereas the MARS hold time produces piecewise-constant allocations. After the mission ends ($t > 152$ s), MARS keeps about 50 PRBs on SC because the risk reserve grows with severity. This persistent reserve explains MARS's lower control-latency proxy (Fig. 8), since the proxy penalizes SC allocations below 35 PRBs. Averaged over 30 seeds, the proxy is the same for both policies while the mission runs (20.9 ms), but it is 16.8 vs. 22.4 ms after the

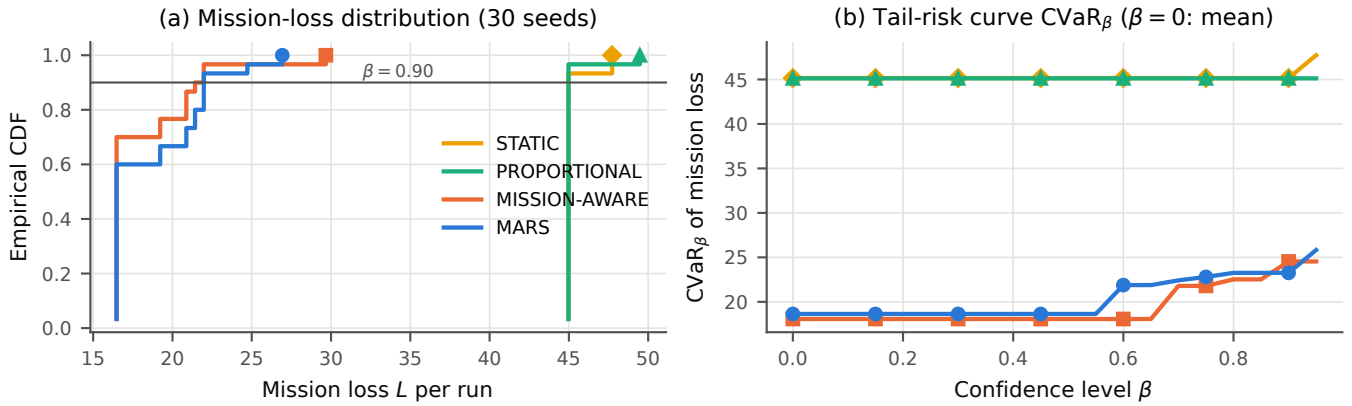


Fig. 4. (a) Empirical CDF of per-run mission loss; markers show the worst run. (b) CVaR$_\beta$ as a function of β; the ranking of MARS and MISSION-AWARE changes with β.

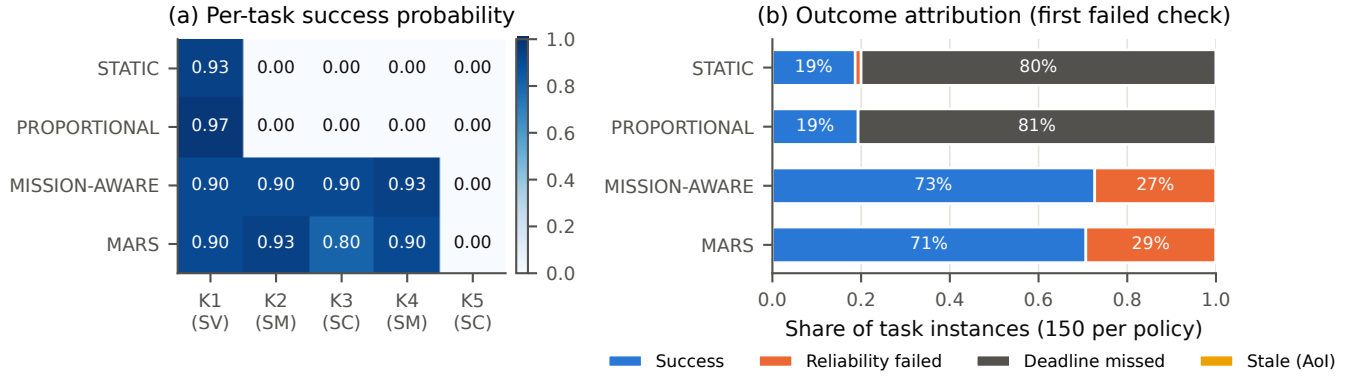


Fig. 5. (a) Per-task success probability (30 seeds). K_5 never succeeds. (b) Outcome of all 150 task instances per policy, attributed to the first failed check (deadline, then reliability, then freshness).

mission ends. The latency advantage therefore comes from the post-mission period and does not reflect better mission service.

VII. Edge-Case Evaluation

A. Radio Budget: Feasibility Floor and Surplus Paradox

Fig. 9 sweeps B from 50 to 200. At $B = 50 = \sum_s B_s^{\min}$ every slice is pinned at its minimum, no policy can adapt, and all success scores are zero. That is the degenerate lower edge. Budgets below 50 violate (1) and are rejected by the configuration validator. Success climbs steeply between 55 and 80 PRBs for the adaptive policies, while STATIC and PROPORTIONAL need about 80 PRBs before even K_1 succeeds.

Counter-intuitively, success *falls* for $B > 120$ and returns to the STATIC level at $B \geq 150$. The cause is the radio–edge coupling of (3). The active SM slice is capped at $B_{\mathrm{SM}}^{\max} = 40$ PRBs, so a larger B shrinks its fraction $40/B$ of the edge capacity. For example, K_2 receives 27% of the edge at $B = 150$ instead of 40% at $B = 100$, becomes compute-bound, and misses its deadline. MARS degrades earlier than MISSION-AWARE (0.60 vs. 0.70 at $B = 120$): under MARS, K_2 misses its deadline in 13% of runs and K_3 fails its reliability check in 47%, against 0% and 27% for MISSION-AWARE. A deployment-grade model should decouple edge scheduling from radio share or scale $B_s^{\max}$ with B.

B. Incident Severity and Network Shocks

Fig. 10 varies σ_0. Under a benign incident ($\sigma_0 \leq 0.2$), MARS reaches 0.96 [0.93, 0.99] versus 0.91 [0.85, 0.96] for MISSION-AWARE at $\sigma_0 = 0$, and 0.95 vs. 0.86 at $\sigma_0 = 0.2$, with markedly lower CVaR (17.2 vs. 21.2). This is the regime in which K_5 is feasible and the protected SC slice pays off. For $\sigma_0 \geq 0.8$ all policies collapse: the adaptive ones to about 0.35, and STATIC and PROPORTIONAL to zero at $\sigma_0 \geq 0.9$, where even K_1 no longer succeeds.

Fig. 11 sweeps the per-step shock probability. With no shocks the two mission-oriented policies coincide (0.78). Success decays smoothly as shocks become more frequent, reaching 0.13–0.16 in a shock storm ($p_{\mathrm{sh}} = 0.3$). MARS's reserve does not give a measurable advantage under frequent shocks.

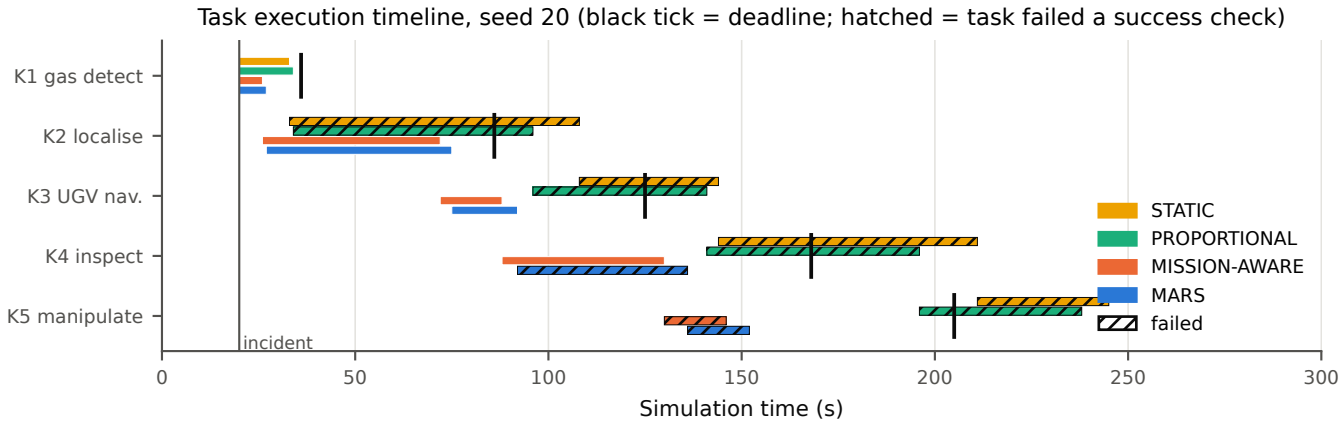


Fig. 6. Task timeline for seed 20. Mission-oriented policies finish the mission by $t \approx 150\,\mathrm{s}$; STATIC and PROPORTIONAL overrun the K_2 deadline and every later deadline.

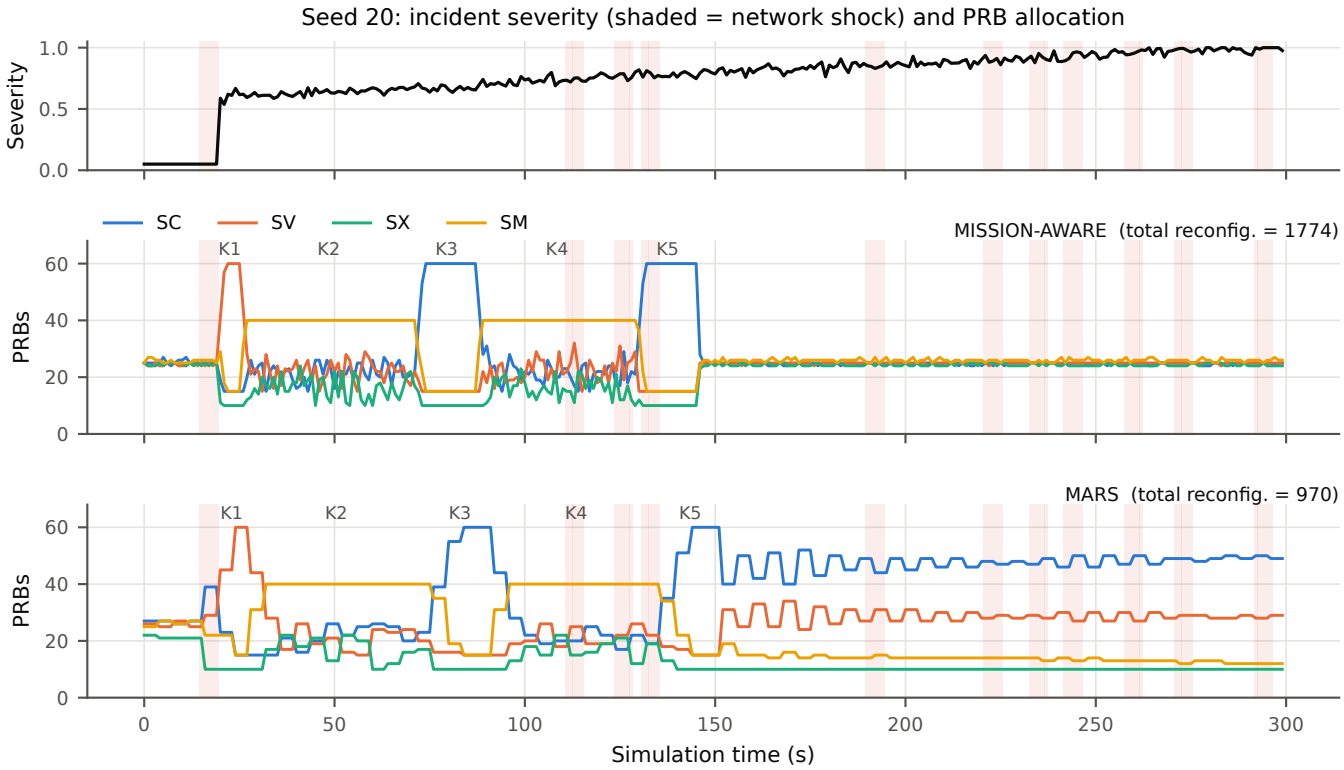


Fig. 7. Seed 20: severity with network shocks (shaded) and PRB allocation under MISSION-AWARE and MARS. Labels mark the active task.

C. Deadline Tightness and Workload

Tightening deadlines (Fig. 12) exposes the cost of hysteresis. At scale 0.8, MISSION-AWARE achieves 0.58 [0.52, 0.63] and MARS 0.49 [0.43, 0.53]; at 0.7 the figures are 0.36 and 0.28. Below 0.6 only K_1 can succeed for either policy. When slack is small, the 4 s hold delays the move of PRBs to a newly activated task, and that delay is no longer recoverable. Relaxed deadlines (scale 1.5) help STATIC and PROPORTIONAL most, since deadlines were their binding constraint.

Workload scaling (Fig. 13) shows the same pattern. At $1.25\times$ load MARS falls to 0.42 vs. 0.58 for MISSION-AWARE. At $3\times$ load the mission no longer fits in the 300 s horizon: only three tasks are observed and the rest count as failures. At light load ($\leq 0.5\times$) the non-adaptive policies recover to 0.4–0.6, so mission awareness matters most when resources are tight relative to workload.

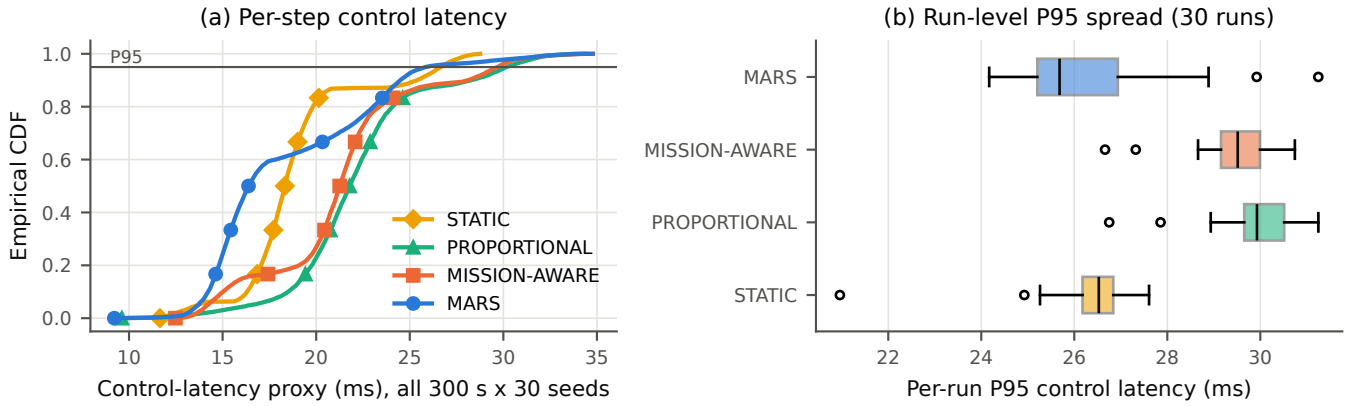


Fig. 8. (a) Per-step control-latency proxy across all runs. (b) Spread of the run-level P95. MARS's low tail arises mainly from its post-mission SC reserve. Values are model proxies, not measured NR latency.

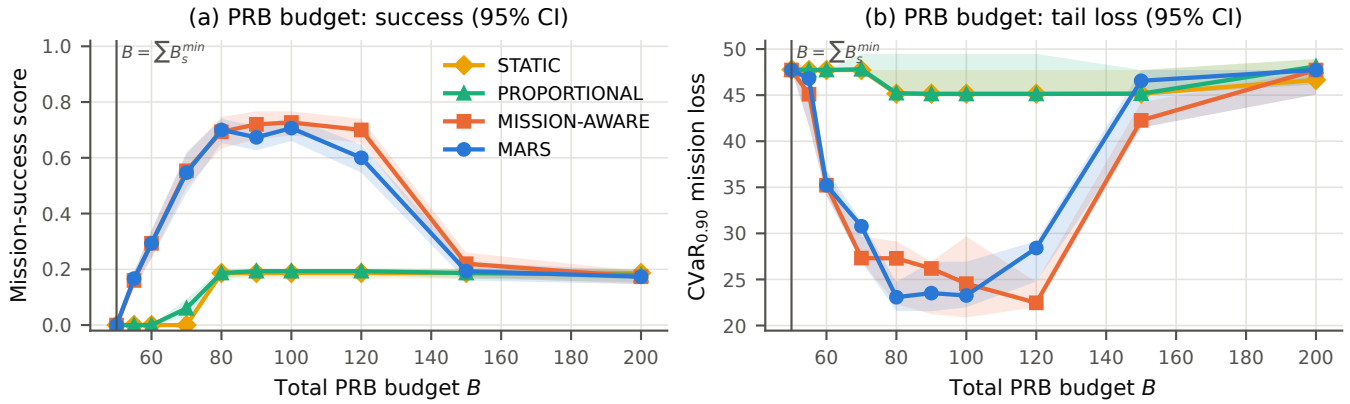


Fig. 9. PRB-budget sweep with 95% CIs. The vertical line marks the feasibility floor $B = \sum_s B_s^{\min} = 50$. Success drops above $B = 120$ because edge share is tied to radio share while the active slice is capped.

D. Joint Stress Map

Fig. 14 crosses severity and budget for all four policies. The mission-oriented policies match or beat the non-adaptive ones in every cell except $B = 120$, $\sigma_0 = 0.8$, where all four policies are near 0.2, and the $B = 60$, $\sigma_0 = 1.0$ corner, where all fail. MARS and MISSION-AWARE differ by at most 0.10 in any cell. The largest gaps are at $B = 100$, $\sigma_0 = 0.2$ (MARS ahead, 0.95 vs. 0.86) and at $B = 120$, $\sigma_0 = 0.6$ (MARS behind because of the budget paradox, 0.60 vs. 0.70).

E. MARS Parameters and Ablation

Fig. 15 sweeps the three MARS parameters. Risk aversion ρ barely moves success (0.68–0.72) and changes reconfiguration cost non-monotonically. The hold time T_h is the main stability lever: going from 0 to 20 s cuts reconfiguration by $9\times$ (2908 to 327) at a cost of 0.08 in success. $T_h = 0$ and $T_h = 1$ behave identically because the hold condition $t - t_{\text{last}} \geq T_h$ is always true for $T_h \leq 1$. A very small change limit ($\Delta_{\max} = 1$) freezes adaptation (success 0.53), and $\Delta_{\max} = 32$ gives the best success (0.73).

The 30-seed ablation (Fig. 16) revises the earlier 6-seed result in three ways. (i) Removing AoI, urgency, or *all* criticality terms yields results identical to MARS-full. (ii) Removing hysteresis *raises* success to 0.727 (it then equals MISSION-AWARE) at a $2.8\times$ higher reconfiguration cost. The earlier 6-seed ablation suggested the opposite. (iii) Removing the risk reserve or the reconfiguration penalty *lowers* reconfiguration cost (to 762 and 746) without hurting success. The penalty measures deviation from the equal share of 25 PRBs, not from the previous allocation, so it pushes allocations toward 25 and away from the bounds where they otherwise settle.

The inert criticality terms are explained by saturation (Fig. 16(c)). Because only one task is executable at a time, $\chi_k \approx 8$–23 dominates the demand terms (< 0.35). While the mission runs on seed 20 ($t = 20$–151 s), the largest softmax weight is at least 0.9997 at every step. The resulting target exceeds the slice's upper bound and is clipped, so the allocation depends only on *which* slice is active, not on *how* critical it is. Making the criticality weights matter requires concurrent tasks or a softmax temperature that keeps scores in the unsaturated range.

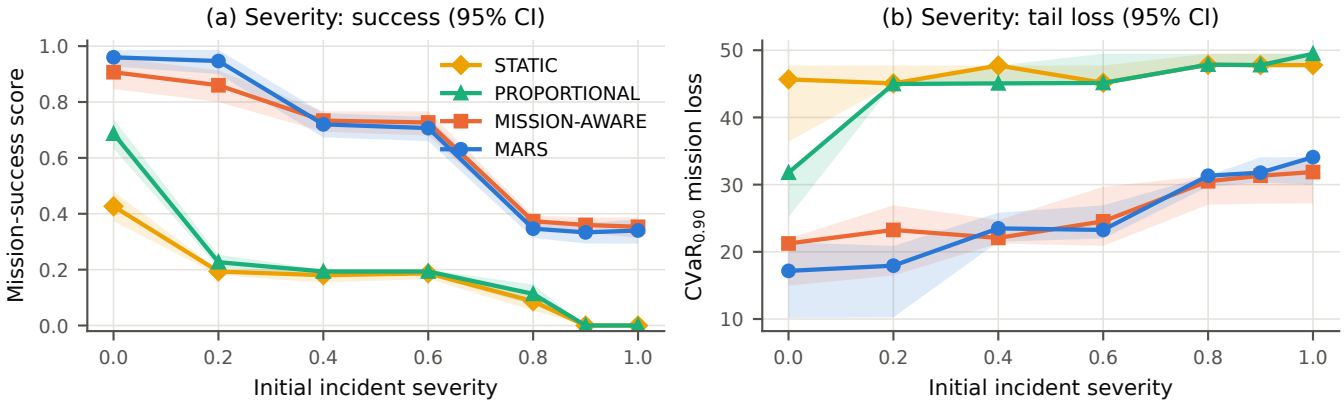


Fig. 10. Initial-severity sweep with 95% CIs. MARS leads at low severity, where the final task becomes feasible; all policies degrade above $\sigma_0 = 0.6$.

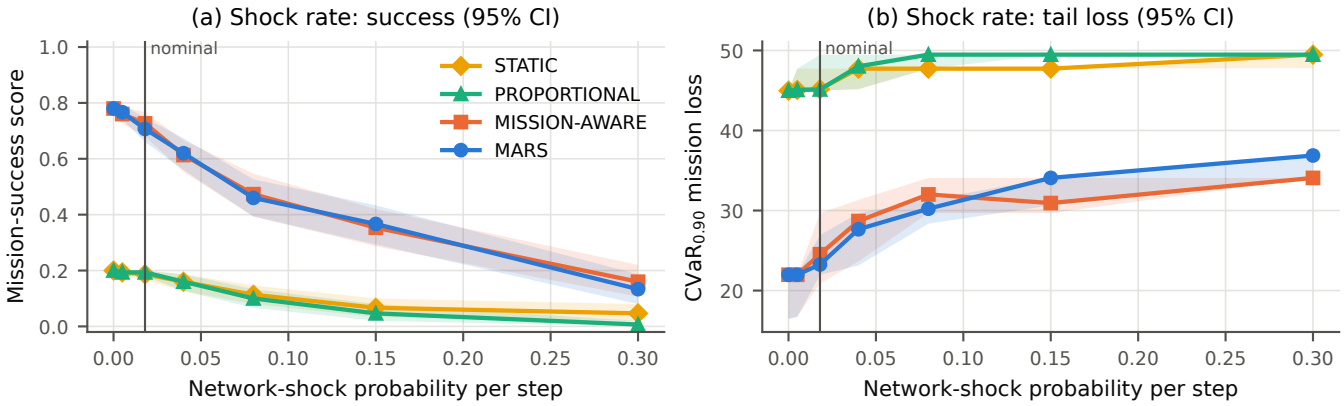


Fig. 11. Network-shock probability sweep with 95% CIs (vertical line = nominal $p_{\mathrm{sh}} = 0.018$).

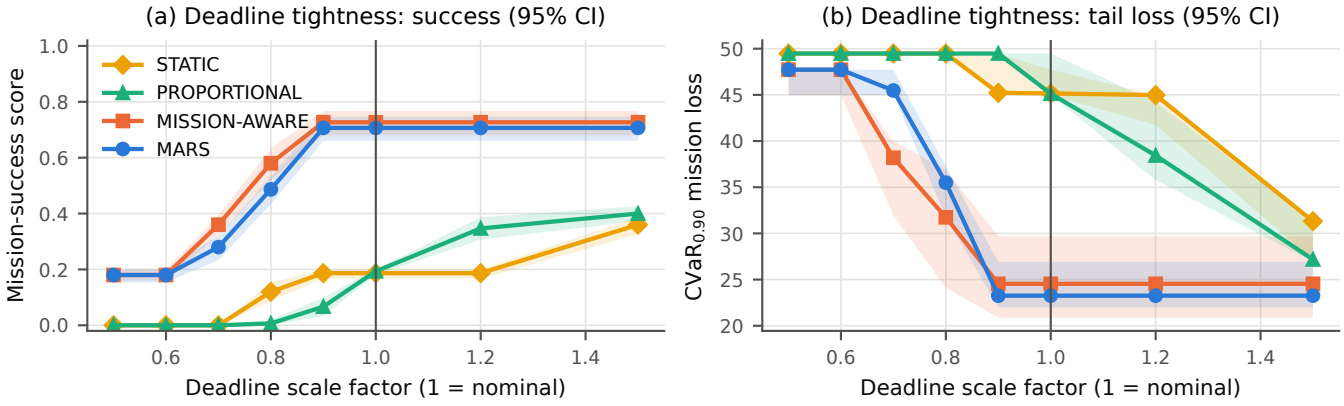


Fig. 12. Deadline-scale sweep. Below 0.6 every policy is limited to K_1; between 0.7 and 0.9 MARS trails MISSION-AWARE because of its hold time.

VIII. Discussion

What the evidence supports. (1) Mission-aware allocation is far better than fixed or demand-proportional slicing in this model. The effect holds in every paired seed and across the whole stress map except degenerate corners. (2) MARS delivers the same nominal mission outcome as MISSION-AWARE with 44% less reconfiguration, which is relevant where each reallocation carries signalling or instability cost. (3) MARS is clearly better under mild incidents.

What it does not support. MARS does not have a statistically supported tail-risk advantage at the nominal point, and its CVaR ranking depends on β. Under tight deadlines,

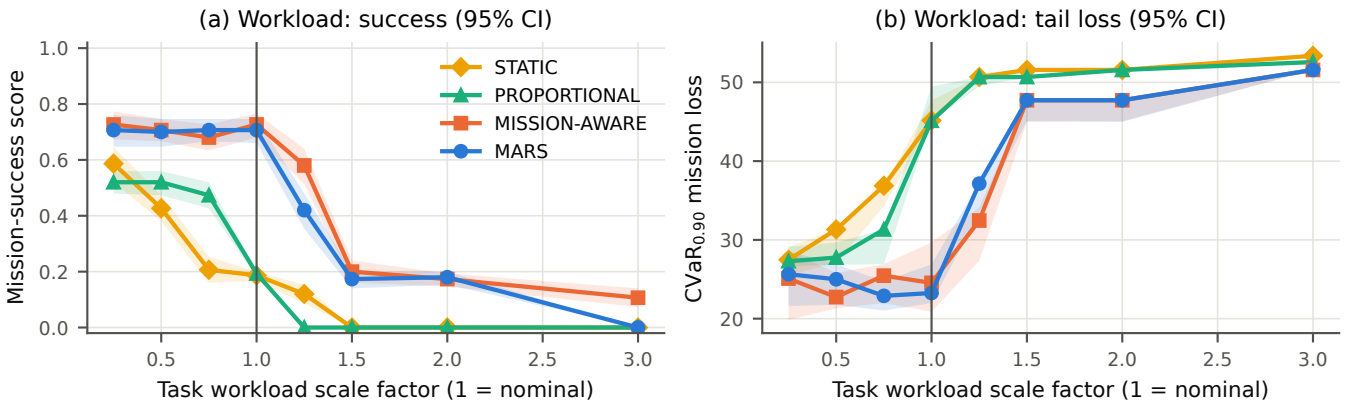


Fig. 13. Workload-scale sweep. Above 1.5× the mission cannot finish within the horizon for any policy.

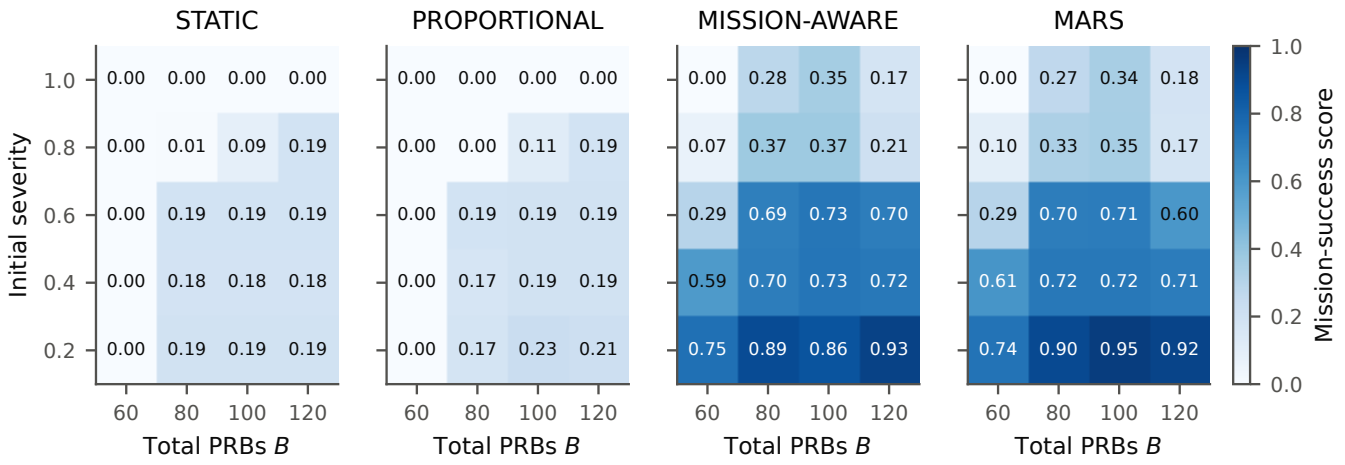


Fig. 14. Mission-success score over initial severity × total PRBs (30 seeds per cell).

heavy workload, or surplus PRBs it is worse than MISSION-AWARE, because the hold time and reserve trade agility for stability. The criticality formulation (7), presented as a core contribution, does not influence decisions in the current single-task-at-a-time configuration.

Design implications. An adaptive hold time that shrinks when deadline slack is small would combine the stability of MARS at nominal load with the agility of MISSION-AWARE under pressure. A reconfiguration penalty relative to $b_s(t-1)$, a reserve that switches off when the mission ends, and an edge scheduler independent of radio share would remove the three model artifacts identified above.

IX. Threats to Validity and Limitations

Internal validity. Results depend on hand-chosen parameters, and several conclusions (e.g., K_5 infeasibility and the budget paradox) follow directly from specific modelling choices that we expose rather than tune away. Thirty seeds give wide $\text{CVaR}_{0.90}$ intervals because only three runs fall in the tail. *Construct validity.* The success score is a per-task fraction, not mission completion. The latency values are proxies that are not part of task success. Mission loss uses value weights $w_k(1+0.25h_k)$ rather than χ_k. *External validity.* The radio model is analytical, with one cell, one edge server, no mobility or handover, and independent component reliabilities. No packet-level, testbed, or field validation has been performed, and no E2/A1 interface is exercised. The controller is a heuristic; it is not compared against an optimization-based or learning-based controller.

Future work should validate the abstractions with ns-3/5G-LENA [10], recalibrate the reliability model so that every task is feasible at nominal severity, introduce concurrent tasks so that criticality weights become decision-relevant, implement the adaptive hold time proposed above, and benchmark against a CVaR-constrained MILP and a reinforcement-learning controller.

X. Conclusion

MARS-O-RAN links a five-task emergency mission to four network slices and evaluates task success, deadlines, freshness, reliability, reconfiguration, and tail loss in a reproducible

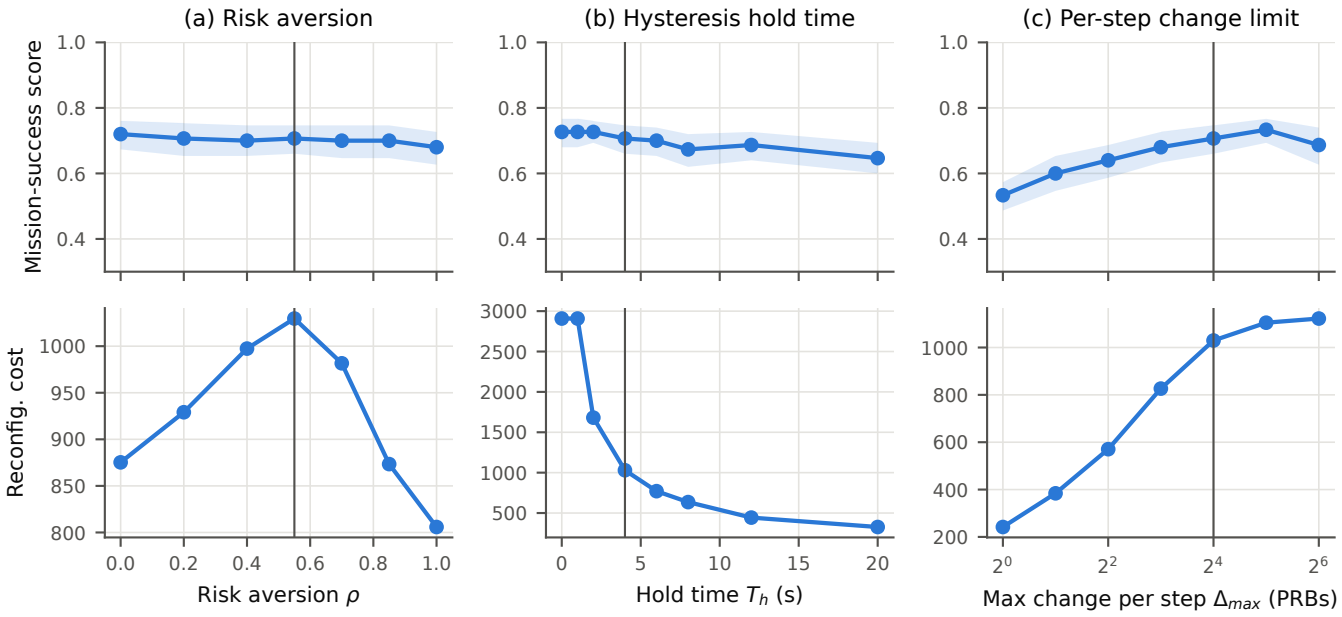


Fig. 15. MARS parameter sensitivity (30 seeds; vertical line = nominal). Top: success with 95% CI. Bottom: reconfiguration cost.

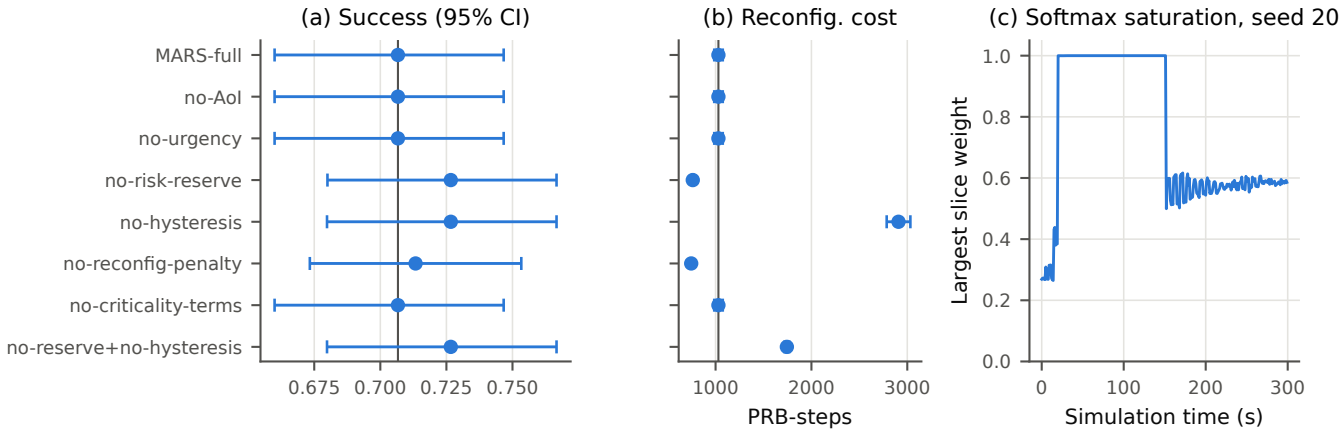


Fig. 16. (a,b) Extended ablation over 30 seeds with 95% CIs (vertical line = MARS-full). AoI, urgency, and all criticality terms have no effect; removing hysteresis increases success but nearly triples reconfiguration. (c) Largest MARS slice weight per step on seed 20: it is saturated at 1 for the whole mission, so slice bounds, not criticality values, set the allocation.

digital twin. A paired statistical analysis and eleven edge-case studies give a more precise picture than aggregate means. Mission-aware slicing is robustly beneficial. The MARS stability mechanisms cut reconfiguration by 44% at equal nominal outcome and help under mild incidents, but they cost agility when slack is tight. The edge cases also uncovered an infeasible final task, a radio–edge budget paradox, and saturated criticality weights. These findings define the next steps toward a controller whose advantages hold across the operating envelope rather than at a single nominal point.

Reproducibility

All results are generated by `src/run_all.py` (main experiment) and `src/edge_case_experiments.py` (statistics, sweeps, and figures) with fixed seeds. Outputs are written to `results/` and `results/extended/`.

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